

The bundle-chain decomposition of a bacterial genome: *Escherichia coli* K-12 at five k -mer lengths

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Abstract

The companion notes computed the sandpile group of a de Bruijn graph on a 5386 bp phage. This one computes it on a 4 641 652 bp bacterium, *Escherichia coli* str. K-12 substr. MG1655 (NC_000913.3), at $k = 31, 51, 75, 100, 150$. The computation is possible because of one observation that costs a line: in the convention used throughout this series every vertex of G_k satisfies $d^+(v) = d^-(v) = \text{mult}(v)$, so a vertex survives unitig contraction *if and only if its k -mer is repeated*. The 4.6-million-vertex graph is therefore never built; the compacted graph is read off the repeated k -mers directly, and the whole pipeline runs in about 13 seconds. Three findings. *One*: the irreducible part of the ambiguity of a 4.6 Mbp genome is a digraph on 115 vertices at $k = 150$ — the chain $4\,641\,652 \rightarrow 23\,933 \rightarrow 115$ — and its group is computed exactly. *Two*: the splitting is extremely lopsided; at $k = 150$ the length-driven summand has order 10^{10439} and the arrangement summand $10^{49.7}$, so 99.5% of the exponent is removable by longer k -mers and 0.5% is removable by no k at all, while the irreducible part still grows by 10^{344} as k falls to 31. *Three*: the seven ribosomal operons do *not* form a multiplicity-seven repeat family, at any k . Five lie on one strand and two on the other, and the single-strand theory of this series sees a repeat only between co-oriented copies; the two minus-strand operons give the longest clean spine in the genome (4030 bp at $k = 150$) and the five plus-strand ones a separate, fragmented family. This is a quantitative measure of what a bidirected theory would have to supply. Six checks, all exact, including a rerun of the whole fast pipeline against the published phage decomposition.

1 Introduction

For a circular sequence S and $k \geq 1$, $G_k(S)$ is the de Bruijn multigraph whose vertices are the distinct k -mers and whose arcs are the occurrences of $(k+1)$ -mers; S is an Eulerian circuit, and by the BEST theorem the number of reconstructions of S from its k -mer spectrum is $|\mathcal{K}(G_k)| \prod_v (d^+(v) - 1)!$ with $\mathcal{K}(G_k)$ the sandpile group [3]. The bundle-chain splitting theorem [5] writes

$$\mathcal{K}(G_k(S)) \cong \bigoplus_{\text{chains } c} (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk}),$$

a *length-driven* summand that longer k -mers remove and an *arrangement* summand, the sandpile group of the repeat-adjacency skeleton, that no k removes. Everything so far has been computed on ϕ X174, 5386 bp. The question this note answers is what the decomposition looks like on a genome three orders of magnitude larger.

What is known. Assemblers have compacted de Bruijn graphs to unitigs for twenty years [10, 8, 1], and the repeat graph is the central object of the long-read assembler Flye [9, 7]; the repeat content of *E. coli* K-12 — seven *rrn* operons, some fifty IS elements in a dozen families, several hundred REP/BIME palindromic elements — has been catalogued since the genome was published [2]. What is new here is not the repeat catalogue but the algebra attached to it: the exact invariant factors of the arrangement group, and the exact split between the part of the ambiguity that a longer read length removes and the part it cannot.

2 Why a 4.6 Mbp genome is tractable

Lemma 1 (Branch vertices are the repeated k -mers). *In $G_k(S)$, $d^+(v) = d^-(v) = \text{mult}(v)$, the number of occurrences of the k -mer v in the circular sequence. Consequently a vertex has in- and out-degree one — i.e. is smoothed away by the unitig contraction of [3, Lem. 4] — if and only if its k -mer occurs exactly once.*

Proof. Each position i contributes exactly one arc, from $S[i..i+k)$ to $S[i+1..i+k+1)$. The arcs leaving v are indexed by the positions at which v occurs, and the arcs entering v by the positions at which v occurs shifted by one; both sets have $\text{mult}(v)$ elements. $\square$

This is immediate, and it is the whole algorithm. The compacted graph is obtained without constructing G_k : let $R = (p_1 < \dots < p_m)$ be the positions at which a repeated k -mer starts. Then the compacted multidigraph has one vertex per distinct repeated k -mer and one arc $S[p_t..p_t+k) \rightarrow S[p_{t+1}..p_{t+1}+k)$ for each t cyclically, because the walk leaving a branch vertex runs through unique k -mers until the next repeat position. For *E. coli* this replaces 4 641 652 vertices by between 23 933 and 34 272 (Table 1), and the decomposition of [5] then runs in one pass, since the chain-step relation is a partial injection [5, Lem. 6].

Finding the repeated k -mers is the only step that touches the whole genome. We hash each of the N positions to 64 bits, take multiplicities with a sort, and then — this is what keeps the result exact — recover the actual k -mer *strings* at the positions whose hash repeats and group by string. A hash collision can therefore only add a candidate that is discarded; it cannot manufacture a repeat. No collision occurred at any k (check (H)).

3 Results

k	distinct k -mers	branch	arcs	chains	skeleton	$\log_{10} \mathcal{K}(\text{Sk}) $	$\log_{10} \mathcal{K}(G_k) $
150	4 594 348	23 933	71 237	107	115	49.71	10 488.89
100	4 588 709	26 631	79 574	166	175	71.11	11 690.57
75	4 584 636	28 579	85 595	243	261	99.80	12 563.08
51	4 579 164	31 011	93 499	342	366	143.17	13 716.83
31	4 570 869	34 272	105 055	683	870	393.75	15 350.54

Table 1: *E. coli* K-12 MG1655 under unitig contraction followed by bundle-chain contraction. “branch” is the number of repeated k -mers (Lemma 1) and equals the vertex count of the compacted graph. All structural entries exact; $\log_{10} |\mathcal{K}(\text{Sk})|$ by float LU, and exact as a group at $k = 150, 100$ (Table 2).

k	$\mathcal{K}(\text{Sk})$, the arrangement group
150	$(\mathbb{Z}/2)^{17} \oplus (\mathbb{Z}/3)^9 \oplus (\mathbb{Z}/4)^{12} \oplus (\mathbb{Z}/5)^{10} \oplus (\mathbb{Z}/7)^3 \oplus (\mathbb{Z}/8)^4 \oplus (\mathbb{Z}/9)^3 \oplus (\mathbb{Z}/16)^3$ $\oplus \mathbb{Z}/27 \oplus (\mathbb{Z}/32)^2 \oplus \mathbb{Z}/49 \oplus \mathbb{Z}/21308303$
100	$(\mathbb{Z}/2)^{29} \oplus (\mathbb{Z}/3)^{16} \oplus (\mathbb{Z}/4)^{15} \oplus (\mathbb{Z}/5)^{15} \oplus (\mathbb{Z}/7)^2 \oplus (\mathbb{Z}/8)^4 \oplus (\mathbb{Z}/9)^2 \oplus (\mathbb{Z}/16)^6$ $\oplus \mathbb{Z}/25 \oplus \mathbb{Z}/27 \oplus (\mathbb{Z}/32)^2 \oplus \mathbb{Z}/49 \oplus \mathbb{Z}/81 \oplus \mathbb{Z}/128 \oplus \mathbb{Z}/512 \oplus \mathbb{Z}/3448649$

Table 2: The arrangement group, exactly, where the skeleton is small enough for the p -adic Smith form (check (K)). At $k = 75, 51, 31$ the skeleton has 261, 366 and 870 vertices and only the order is given; a 100- to 400-digit determinant could not be factored in any case.

Remark 2 (The shape of the answer). Three things are worth reading off Table 1.

The skeleton is small. At $k = 150$ the entire arrangement ambiguity of a 4.6 Mbp genome is carried by a digraph on 115 vertices: $4\,641\,652 \rightarrow 23\,933$ by unitig contraction, $23\,933 \rightarrow 115$ by bundle-chain contraction. The second step is the one the theory supplies, and it is the larger reduction in proportion.

The split is lopsided. At $k = 150$ the length-driven summand has order $10^{10439.18}$ and the arrangement summand $10^{49.71}$: 99.53% of the exponent of $|\mathcal{K}(G_{150})|$ is removable by reading longer k -mers, and 0.47% is not removable by any k . The percentage is not the interesting part — $10^{49.7}$ is still an enormous number of globally distinct threadings, and it is the part that only long reads, mate pairs or a reference can resolve.

The irreducible part grows fast as k falls. From $k = 150$ to $k = 31$ the arrangement summand grows from $10^{49.71}$ to $10^{393.75}$, a factor of 10^{344} , while the skeleton grows from 115 to 870 vertices. Shortening the read length does not merely leave more length-driven ambiguity; it creates genuinely new arrangement ambiguity, because repeats shorter than the old k enter the graph and connect parts of the skeleton that were previously unrelated.

By the BEST theorem the number of circular sequences with exactly the k -mer spectrum of this genome is $|\mathcal{K}(G_k)| \prod_v (d^+(v) - 1)!$, which is $10^{20735.14}$ at $k = 150$ and $10^{31748.76}$ at $k = 31$.

4 The ribosomal operons, and a correction

E. coli K-12 has seven ribosomal RNA operons, each about 5 kb, and the natural expectation — the one the task that prompted this note recorded — is a repeat family of multiplicity seven. *No such family exists, at any k .*

The reason is structural, not numerical. The NCBI feature table places five operons on the + strand (at 223 771, 3 941 808, 4 035 531, 4 166 659, 4 208 147) and two on the – strand (at 2 726 069 and 3 423 423). The theory of [3, 4, 5, 6] is single-strand: two copies are a repeat of $G_k(S)$ only if they are co-oriented *in* S . The seven operons therefore cannot form one family. They split by strand, and the computation confirms the split exactly (check (R)):

- the **longest bundle chain in the whole genome**, at every k tested, is the multiplicity-*two* spine of the two minus-strand operons — $s = 3881$ at $k = 150$ (span 4030 bp), $s = 4412$ at $k = 51$ (span 4462 bp) — annotated 16S, 23S and 5S ribosomal RNA. Two copies annotated on the same (minus) strand are co-oriented with each other, hence a direct repeat in S ;
- the five plus-strand operons give the multiplicity-*five* family, whose longest chain is the 16S gene (span 775 bp), with copies at 223 997, 3 942 034, 4 035 757, 4 166 885 and 4 208 373 — all five operons;

- the five are not identical to one another either, so the rRNA region additionally throws off chains of multiplicity three and four, one per subset of copies that agree.

Multiplicity seven does occur in this genome, but it is IS5A, not rRNA (five chains, 88 k -mers at $k = 31$), as are multiplicities eight and nine. The other long chains are all insertion sequences: IS186A ($\times 3$, 1345 bp), IS2D ($\times 3$, 1331 bp), IS3B ($\times 2$, 1260 bp), IS3A ($\times 3$, 1255 bp), IS5U ($\times 3$, 1202 bp). At $k = 31$ the highest multiplicities are 22 and 23 with spans of 32 to 35 bp: the REP/BIME extragenic palindromic elements, which are shorter than 51 and so vanish completely from the graph at every larger k tested.

Remark 3 (What this measures). A bidirected theory — the open problem of [4, Open (a)] — would see all seven operons as one family, and would see the IS copies that sit in inverted orientation. The $5 + 2$ split is a quantitative statement of what the single-strand restriction costs on a real genome: not a footnote, but the dominant feature of the largest repeat family present. We do not have the bidirected theory and claim nothing about what it would give; we can now say precisely what it would have to explain.

5 Verification

The script `ecoli_sandpile.py` runs the six checks of Table 3 in about 13 seconds under half a gigabyte; its output is archived as `ecoli_sandpile_output.txt`. The genome and the feature table are cached beside it and re-fetched from the NCBI E-utilities API if absent.

key	check	mode
(V)	the fast path reproduces the published φ X174 decomposition, $k = 9, \dots, 12$	exact
(H)	hashing finds candidates, strings decide; no collision survives	exact
(C)	the compacted graph is balanced with $d^+(v) = \text{mult}(v)$	exact
(B)	maximal bundle chains in one pass; Lemma 6 injectivity asserted on 1541 chains	exact
(K)	the arrangement group, exactly for skeletons of ≤ 200 vertices	exact/numeric
(R)	no rRNA chain has multiplicity 7; the longest chain is the minus-strand pair	exact

Table 3: The verification battery, 6/6.

Check (V) deserves a word. The pipeline used here is not the one used in [5]: it never builds G_k , it contracts chains in a single pass instead of iteratively, and it takes the Smith form p -adically instead of by Euclidean elimination. Run on φ X174 it returns, at $k = 9, 10, 11, 12$, the same branch counts (2, 10, 47, 146), chain counts (1, 2, 8, 28), skeleton sizes (1, 8, 37, 110), length-driven exponents and the same exact skeleton groups — including $\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$ at $k = 10$ and the seventeen elementary divisors at $k = 9$. Three independent implementations of the same theorem agreeing on a genome is the only protection available when the target genome is too large to check by any other route.

6 Scope

Exact. Every structural quantity — repeated k -mers, branch counts, the multiset of (r_c, ℓ_c) , skeleton sizes, genomic coordinates, the length-driven summand — is exact integer work, as is $\mathcal{K}(\text{Sk})$ at $k = 150$ and $k = 100$ and the location of every repeat family against the feature table.

Numeric. $\log_{10} |\mathcal{K}(\text{Sk})|$ at $k = 75, 51, 31$ is a float LU determinant (`slogdet`) of a 260- to 869-dimensional matrix; the relative error in the logarithm is at the level of machine precision times the dimension, but it is not an exact integer and is not claimed to be.

Single-strand. As in [3, 4, 5, 6]. Section 4 is the measurement of what that costs here.

Open. (a) The two exact arrangement groups each carry one large prime factor (21 308 303 at $k = 150$, 3 448 649 at $k = 100$) that we do not interpret; as in [5] that is what the sandpile group of a generic digraph looks like, and no topological reading is claimed. (b) The 870-vertex skeleton at $k = 31$ is out of exact reach; a modular Smith form would settle it, and nothing in principle prevents that. (c) Whether the 10^{344} growth of the arrangement summand between $k = 150$ and $k = 31$ has a law in terms of the repeat-length distribution.

Provenance

Unrefereed preprint. Lemma 1 is elementary and is stated because it is what makes the computation possible, not because it is deep; everything else in this note is computation on public data. The genome is NC_000913.3 and the annotation is the NCBI feature table for the same accession, both downloaded by the script and cached with it. The expectation of a multiplicity-seven rRNA family, which Section 4 corrects, was the premise of the task that prompted this note, and is recorded here because the correction is the useful part.

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